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THE 50TH RICHARD G. SCOBEE MEMORIAL LECTURE

Phoria Adaptation: The Ghost in the Machine

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ABSTRACT

Importance: Phoria adaptation is a central tonus mechanism that provides plasticity to binocular alignment. However, its slow dissipation inevitably adulterates our clinical strabismus measurements.
Purpose: To examine the role of phoria adaptation in normal binocular control, understand its neural substrate, and explore how it can alter our clinical measurements in common forms of strabismus.
Methods: Investigation into the role of phoria adaptation in maintaining binocular alignment and its role in altering clinical strabismus measurement.
Results: Phoria adaptation permeates all aspects of binocular alignment. It accounts for the stability of orthophoria, "latent" phorias, tenacious proximal fusion in intermittent exotropia, large fusional vertical amplitudes in congenital superior oblique muscle palsy, the "eating up" of prisms in accommodative esotropia, the smaller measured distance deviation in patients with high AC/A ratio or convergence excess, absence of physiologic skew deviation during head tilt, fusional divergence amplitudes, and spread of comitance. This binocular control system probably arises from a cerebellar learning mechanism that involves input via climbing fibers to the inferior olive, which provide a powerful timing and error signal to the cerebellar Purkinje cells to produce activity-dependent modification analogous to long-term potentiation.
Conclusions: Phoria adaptation is generated by a central neural integrator that provides inertia, plasticity, and positional stability to human binocular vision.

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KEYWORDS
Phoria adaptation; normal binocular control; strabismus

"*The important factor concerning fusional movements is not the movement that leads to sensory fusion per se but the prolonged tonic fundamental innervation of the eye muscles that is adapted for the conditions of single vision.*"[1]

Introduction

I esteem it a great honor to be invited to give the 50th Scobee Lecture here in my hometown of San Francisco. Dr. Richard Scobee was revered as a consummate teacher, clinician, and diagnostician. His teachings permeate our field to such a degree that we don't always recognize them in the fabric of what we do. Dr. Scobee served as a founding member of the American Orthoptic Council and founded the American Orthoptic Journal, which was first published in 1951, just a year before his untimely death. His definitive textbook, *The Oculorotary Muscles*,[2] provided the foundation for the field of orthoptics, and created a cogent framework for what was to become an emerging field of study. So it was nice that, while preparing this lecture, so many of Dr. Scobee's seminal observations seemed to resurface within a new context. I hope that my choice of topics helps to carry the torch and honor his memory.

A *phoria* refers to any deviation of the eyes from one of alignment (orthophoria) when binocular visual input is preempted.[3] Strabismus occurs when a phoria persists under binocular conditions, resulting in a manifest deviation (a *tropia*). Strabismus surgery is predicated on the notion that alternate occlusion unmasks the underlying phoria. By targeting surgical dosing to the measured phoria, the resting position of binocular alignment can be reset to near zero, allowing fusion to actively maintain binocular alignment. As pointed out by Lancaster, it is tacitly assumed that the resting deviation when the fusion reflex is preempted (i.e. under conditions of monocular fixation) is identical to the resting deviation when the fixation reflex is prevented in both eyes. A recent study has shown this to be the case for humans with normal binocular vision (Brodsky).[4]

Despite the fact that our eyes stay straight, binocular vision is not a static function.[5] Maintaining binocular vision is a dynamic force, controlled by both momentary

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peripheral fusion (termed *motor fusion*), tonic vergence, and vergence adaptation (termed *phoria adaptation*) that provides the necessary neural integration to minimize the baseline phoria and thereby stabilize binocular vision. Changes in vergence consist of a fast phasic response followed by a slow tonic response. Alternate occlusion represents the elimination of fast fusional vergences, which must then be followed by a longer decay of the slow fusional vergence response.[6] A prism bar measures step-like changes which is more reflective of the rapid phasic response while synoptophore testing measures the slow tonic response.

What is phoria adaptation?

Phoria adaptation is the slow buildup of tonic vergence innervation to establish and sustain a new vergence set point following placement of a prism over one or both eyes.[7–10] Phoria adaptation recalibrates extraocular muscle tonus to realign the visual axes and thereby meet the demands of binocular alignment,[11] and restore the dynamic range (or fusional reserve) in which fast fusional vergence can function.[12] Phoria adaptation resets binocular alignment toward orthophoria; thereby compensating for developmental, environmental, or pathologic alterations in the binocular mechanism.[12] Because phoria adaptation is stimulated by *the effort put forth* by the fast fusional vergence system, repeated tests of fusional amplitude will increase the limits of fusional movement.[13,14] After removal of a prism, the rate of disappearance of this change in tonus is inversely proportional to the length of time this tonicity is maintained.[6,8,13,14]

Clinical investigation into phoria adaptation as a malleable but invisible force encompasses the history of pediatric ophthalmology, capturing the interest of many progenitors of our field such as Hering,[15] Maddox,[16] Hofmann and Bielschowsky,[1,17] Ogle,[18,19] Reinecke,[10] and Guyton.[20] Its slow time decay was characterized by Hofmann and Bielschowsky over a century ago.[1,6,9,20] Phoria adaptation provides an innervational repository to modulate vergence tonus to stabilize horizontal, vertical, and torsional binocular alignment.[21]

Phoria adaptation serves to establish a new equilibrium as the starting position for further vergence eye movements and thus decreases the effort for either convergence or divergence eye movements.[22] This resiliency and tenacity means that a measured phoria can change transiently following active vergence effort.[5,13,14] Because of its lingering effects, however, phoria adaptation can alter measurements of both vergence ranges (which reflect activity of both the fast and the slow vergence system) and also of vergence facility (which examine the patient's ability to shift rapidly between divergence and convergence as a function of the fast fusional vergence system).[7,23] For this reason, phorias are routinely measured before vergence amplitudes are measured.[6] Although phoria adaptation is classically elicited by introducing prisms before one eye to induce a vergence error, this compensatory mechanism can insinuate itself into many forms of intermittent strabismus. Because phoria adaptation has a *memory*, its slow dissipation serves to mask or conceal a large phoria on clinical examination.[17]

Phoria adaptation operates independently of the fast vergence system and thereby minimizes the fixation disparity that results in the need for vergence control.[9,18] Ogle and Prangen found that an increase in slow fusional vergence was accompanied by a reciprocal reduction in fast fusional vergence.[19] However, these two complementary systems may exist on a spectrum wherein some individuals who use disparity as a primary driver of fusional vergence show larger phorias and less inherent capacity for phoria adaptation, while others show strong phoria adaptation and minimal disparity-driven vergence (*written communication, Anna Horwood, July 20, 2017*).

Stimulus for phoria adaptation

Binocular disparity does not generate phoria adaptation. There is always binocular disparity in a complex world (in different planes even when the eyes are aligned), and this binocular disparity is a necessary stimulus for stereopsis. While binocular retinal image disparity provides the stimulus for the fast fusional vergence system, it is the *motor signal* from the fast fusional vergence system that serves as the error signal for phoria adaptation.[5] The input to the slow neural integrator for phoria adaptation is generated by *the output or effort* of fast fusional vergence. Because there has to be an effort to fuse for phoria adaptation to be recruited, fixation disparity induced by a conflict in vergence and accommodation may provide a driving stimulus for prism-induced phoria adaptation[9] (Figure 1). The fast vergence system has a neural integrator with a short half-life of 10–15 s.[7,14] By contrast, phoria adaptation consists of a more stable neural integrator that leaks with a much longer time constant (varying from minutes to several days).[5,9] The output of the slow neural integrator allows for a reciprocal reduction of the output of fast fusional vergence by means of the negative feedback loop in which the system is trying to adapt until the phoria is controlling the entire response.[9] As the slow fusional vergence system charges up, the fast fusional vergence system decays to maintain a constant total output.[6,7] This feedback loop presumably reduces the fatigue and eyestrain that would result from the prolonged use of the fast vergence system.[7–9]

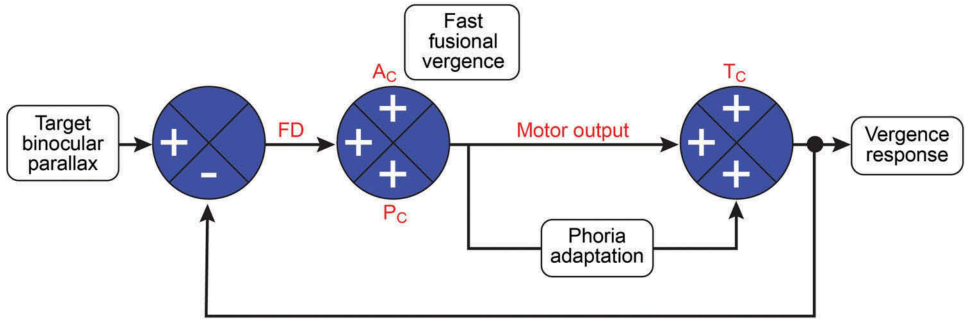

Figure 1. Diagram depicting feedback interaction between slow and fast fusional vergence. The fast fusional vergence system uses binocular retinal image disparity as its input signal. Phoria adaptation uses *the motor output of the fast fusional vergence system* as its error signal to reset tonic vergence and reduce fixation disparity. The output of this slow neural integrator then allows for reciprocal reduction of fast fusional vergence by means of a negative feedback loop to keep the final innervational output constant.

Measurement of phoria adaptation

Phoria adaptation can be estimated but not precisely quantified, due in part to the fact that it is slower to dissipate after an extended period of fusional demands.[6,9,13,17] Measurement of fusional vergence amplitudes using a prism bar placed before one eye under binocular conditions and prolonged monocular occlusion are two examination techniques that help to uncover the degree to which phoria adaptation is operative.[9] Depending on the preexisting level of phoria adaptation, prism bar testing can serve to induce or dissipate phoria adaptation (a critical distinction not determinable by the examiner), whereas prolonged occlusion removes the input of disparity vergence and presumably dissipates any preexisting phoria adaptation. Although monocular occlusion may dissipate phoria adaptation, it remains unclear whether prism bars and monocular occlusion operate by equivalent mechanisms and produce similar endpoints, since monocular occlusion seems to exert other effects that go beyond disrupting fusional vergence and dissipating phoria adaptation (similar to those of Marlow occlusion).[6,24] Monocular occlusion may also disrupt other adaptive processes that have different time constants.

Resilience of phoria adaptation

In patients with intermittent strabismus, phoria adaptation seems to operate most effectively at fixation distances that place binocular alignment within its range.[25] It can also correct different degrees of deviation in different fields of gaze.[20,21,25,26] This phenomenon explains patients ready adjustment to the prismatic effects of anisometropic spectacles in different fields of gaze.[6,27–30] The finding that noncomitant vertical phoria adaptation dissipates much more quickly than comitant phoria adaptation (31 min versus 83 min),[31] demonstrates the coexistence of both global and local mechanisms that subserve phoria adaptation, each having different capabilities and time courses of action.[5,26]

Time course of phoria adaptation

Phoria adaptation begins to initiate quickly but builds and dissipates more slowly.[1,6,9] Even brief presentations of small disparities may be enough to alter phoria adaptation.[14] Estimates for initiation have ranged from as little as 100 ms to 30 min of binocular visual experience.[9,32,33] These wide ranges probably reflect some degree of immediate initiation followed by a slower build-up period.[7] According to Henson and North, phoria adaptation is substantially complete after 2–3 min of binocular visual experience.[33] An earlier study by Ogle et al. had found that vertical prism adaptation was complete within 3–7 min and that adaptation followed a logarithmic curve.[18] A larger prism takes longer to adapt to but does not increase adaptation time in a linear manner.[7]

However, the rate at which phoria adaptation dissipates varies considerably, ranging from 30 s in some individuals to several days in others.[7] This measured variability may in part reflect the inherently different time constants for phoria adaptation between individuals, as well as the inverse correlation between the degree of phoria adaptation and the duration of the previous vergence effort.[7,13,34] Rosenfield et al. found that 20 min of monocular occlusion was sufficient to estimate the latent deviation.[35] However, a longstanding phoria adaptation may require hours to fully dissipate.[5,31,36–38]

Protean clinical manifestations

Phoria adaptation infuses our binocular alignment with a kind of inertia,[13] which can corrupt or override our clinical measurements. Conversely, when phoria adaptation is fully operational, the phoria that we measure with prism and alternate cover testing can be spuriously low. Prolonged patching often increases the measured deviation by breaking down phoria adaptation to reset binocular misalignment toward its uncompensated value.

Phoria adaptation comes in many guises, with clinical effects that are much more protean than generally appreciated. What follows are some examples of conditions in which alternate cover testing of the phoria measurement fails us when the superimposed effects of phoria adaptation are not recognized.

1) **Stillness**: Over a century ago, Sherrington remarked that "The role of muscle as an executant of movements is so striking that its office in preventing movement and displacement is somewhat overlooked."[39] An overlooked but essential quality of phoria adaptation is the *stillness* that it confers upon our binocular alignment. While friction accounts for the stillness of many inanimate objects, stillness is a formidable task for biological systems. As articulated by Carpenter: "Why do we take stillness so much for granted? … That while a muscle is maintaining a constant length and thus doing no work, the actin-myosin bonds between the sliding filaments are ceaselessly breaking and reforming, consuming ATP and so wastefully turning precious energy into heat rather than work. To maintain a stationary gaze demands that the brain sends continual commands to the contractile fibers in the ocular muscles and poses as many problems as moving the eyes fast. Thus, stillness is not an easy task."[40] In fact, biological systems are never truly still, as evidenced by the tiny fixational intrusions which promote stability of visual perception via the Troxler phenomenon.[41] Phoria adaptation also explains the recent observation that the brain does not require visual input to maintain binocular alignment in the dark on a short-term basis.[4]

2) **Orthophorization**: Despite the baseline divergent anatomical position of the eyes within the orbits, normal humans have little or no measurable horizontal phoria on alternate cover testing.[10] Phoria adaptation provides the interstitial force to maintain approximate orthophoria, which explains why "distance phorias are not normally distributed but instead showed abnormally high clustering near orthophoria prior to an extended period of monocular occlusion."[6] An extended period of reading prior to a phoria measurement may generate a more convergent phoria.[14]

3) "**Latent" phoria**: When measuring a binocular deviation with progressive prisms, neutralization frequently occurs long before the deviation can be reversed with a stronger prism. This finding of a "latent phoria" probably reflects the hidden presence of phoria adaptation.[1,6,14] It is therefore customary to assign the value of the full deviation to the maximum measured angle that produces no reversal of the initial deviation. When we say "This patient has a small intermittent exotropia that slowly builds to 35 prism diopters," we are tacitly acknowledging the effects of phoria adaptation.

4) **Accommodative esotropia**: Phoria adaptation may mask the true esodeviation in children with accommodative esotropia, who are known to "eat up prisms," meaning that when temporary prisms are placed on top of the full cycloplegic refraction to neutralize the esodeviation, a portion of the esodeviation will slowly recur through the prismatic correction. Accordingly, surgery based on the initial measurements may lead to undercorrection.[42] In this setting, it is unknown whether phoria adaptation unmasks the full esodeviation or whether it actively generates a larger esodeviation.[43] There is further controversy as to whether the robust prism adaptation response in accommodative esotropia reflects or requires the presence of anomalous retinal correspondence.[44]

Some children with accommodative esotropia have a greater esodeviation at near, associated with a high AC/A ratio or convergence excess. Nevertheless, strabismus surgery targeted to the larger near esodeviation does not produce surgical overcorrection of the distance esodeviation. This finding suggests that the measured distance deviation is artificially low, a finding which implicates phoria adaptation.[6,12] The recent finding that both prism adaptation and prolonged monocular occlusion can increase the distance deviation further supports this notion.[45,46] So phoria adaptation may explain why you can target surgery for the near deviation and not overcorrect for distance.

Although phoria adaptation is a disparity-driven response, vergence aftereffects can also adapt to accommodative vergence (and vice versa), demonstrating a versatile complementary crosslinking and plasticity between the tonic vergence and accommodative systems to match changing demands[47–49] (Figure 2). Thus, in some people (like those with accommodative esotropia), accommodation drives vergence to produce an AC/A ratio while in others (like those with intermittent exotropia), convergence drives accommodation to produce a CA/C ratio.[12]

Phoria adaptation can sometimes result from sustaining a vergence effort for as little as 30 s.[5] "Cellphone vision," which follows prolonged reading of small font at close range, causes distant objects to appear distorted (not quite blurred but not quite double) for several minutes. This symptom may be driven by the transitory effects of phoria adaptation, which throws the vergence system into disarray by causing an esophoria, perhaps contributed to by the secondary effects of tonic accommodation.[49]

5) **Intermittent exotropia**: Patients with intermittent exotropia display a greater distance than near exodeviation on initial alternate cover testing. It was Scobee who noted that the near exodeviation usually increases to approximate the distance exodeviation after 45 min of monocular occlusion.[2] This "tenacious proximal fusion"[50] reflects the role of phoria adaptation, which impedes expression of the full near exodeviation.[12] Although it is generally believed the presence of large convergence amplitudes are necessary to control intermittent exotropia, it may be that phoria adaptation serves this function,[12] which would explain why correlative fusional convergence amplitudes (i.e. equal to the amplitude of the exodeviation plus superimposed convergence amplitudes) are no longer detectable immediately following surgical correction of the exodeviation (Figure 2). So phoria adaptation also explains why we can target surgery for the distance deviation and not overcorrect the near deviation (except in the rare patients who have a true high AC/A ratio).

Prism adaptation or prolonged monocular occlusion can also increase the measured angle in patients with intermittent exotropia and more accurately direct surgical dosing.[51–53] However, its use as a stand-alone treatment to reduce hemiretinal suppression and restore binocular control in intermittent exotropia has never met with success, perhaps owing to the correlative dissipation of phoria adaptation that ensues. While some patients with intermittent exotropia immediately break down to their full deviation (indicating control by fast vergence), others break down slowly with prism and alternate cover testing, suggesting that phoria adaptation is being used to control the deviation. The prognosis for relative postoperative control in these two overlapping groups is unknown.

6) **Congenital superior oblique palsy**: Patients with congenital superior oblique palsy are said to have "increased fusional vergence amplitudes," meaning that these patients will "eat up prisms" and show a large induced vertical deviation when a base down prism bar is increased progressively before the paretic eye. This stepwise dissipation of fusional control to yield a measured deviation far beyond the normal range of vertical divergence probably reflects the slow dissipation of phoria adaptation.[6,54] As with intermittent exotropia, these "increased vertical fusional amplitudes" disappear immediately following successful strabismus surgery, signifying that phoria adaptation rather than peripheral fusion was masking the true hyperdeviation.

7) **Postoperative prism adaptation**: When performing strabismus surgery in adults who have worn prisms to treat diplopia, we admonish them to discontinue their prism glasses postoperatively.[6,8] We do this not only to prevent postoperative diplopia, but to prevent

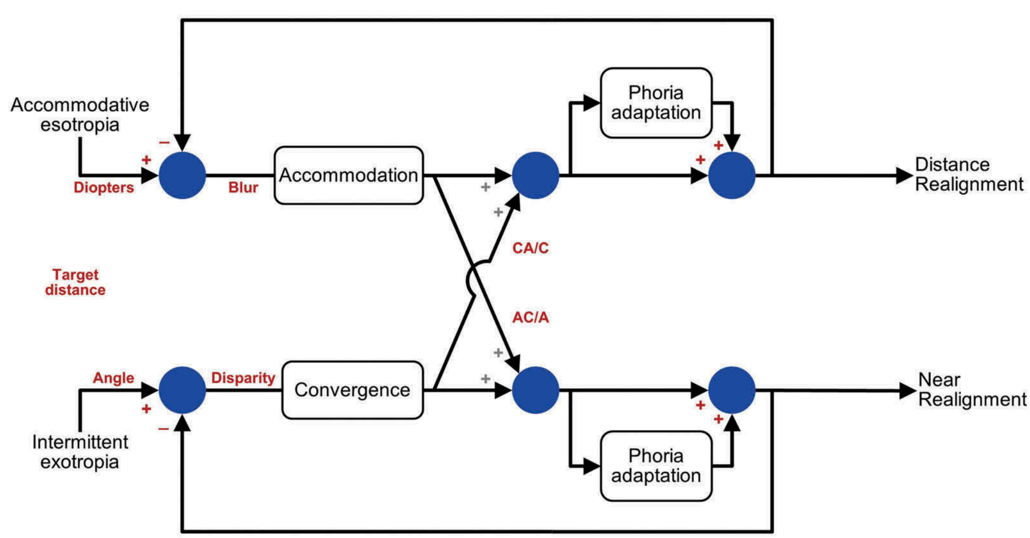

Figure 2. Diagram depicting the general mechanism of phoria adaptation as it influences vergence–accommodation interaction. Blur-driven accommodation implements a similar neural integrator as does disparity-driven vergence to program tonicity. Note the cross-linking that generates accommodation-linked convergence (AC/A) and convergence-linked accommodation (CA/C). For both convergence and for the accommodative system, the fast system provides the immediate phasic response to a change in disparity or blur. The tonic adaptation uses the motor output of the fast system to provide a slower adjustment in the tonic level of accommodation or vergence. In both systems, phoria adaptation is driven by the motor signal and not by sensory input to the motor signal. And in some people (like those with accommodative esotropia), accommodation drives vergence to produce an AC/A ratio while in others like those with intermittent exotropia, convergence drives accommodation to produce a CA/C ratio. Phoria adaptation explains why strabismus surgery targeted for the distance deviation does not overcorrect the near deviation in intermittent exotropia, and why strabismus surgery targeted to the near deviation does not overcorrect the distance deviation in accommodative esotropia (reprinted from ref 12, with permission).

a prism-induced recurrence of the deviation. Instructing patients not to wear their prisms after strabismus surgery betrays an unconscious awareness of phoria adaptation.

8) **Physiologic skew deviation**: Despite the predicted effects of otolithic input to the extraocular muscles, normal binocular individuals have no measurable hyperphoria during head tilt to either side.[55] Phoria adaptation may provide the “antivestibular” force that maintains orthophoria during head tilt. Despite its slow decay, it retains a fluidity which allows it to kick in to different degrees in different fields of gaze. This is not surprising since the vestibular system and the binocular system each rely upon a central neural integrator with a slow decay (velocity storage for the vestibular system and phoria adaptation for the eyes) to produce a perseverative signal that stabilizes balance and perception. This interconnection may explain why phoria adaptation can modify vestibulo-ocular adaptation[56] and vice versa.[5,57] As pointed out to me by Cameron Parsa, MD, and Stephen Archer, MD, many patients have tiny amounts of skew deviation (on the order of 1–2 PD) yet they are unable to fuse it, and they do not prism adapt when a vertical prism is used to eliminate the diplopia. So the central vestibular pathways that generate skew deviation may also be integral to the pathogenesis of phoria adaptation.

9) **Fusional vergence amplitudes**: As divined long ago by Ogle et al., phoria adaptation is not a measure of fusional reserves, since the same fusional reserves are measurable before and after adaptation.[18] Strong evidence for this mechanism for phoria adaptation is evident in patients with intermittent exotropia as well as those with congenital superior oblique palsy. Phoria adaptation can create the false appearance of large fusional vergence amplitudes when it masks a latent phoria.[14] Phoria adaptation explains why the measured divergence break and recovery points are lower when tested after convergence.[7,23,58–60] It also explains Scobee’s observation that fusional vergence amplitudes, as measured at near, seem to correlate with the horizontal phoria measurement as measured at distance.[61] Scobee found that patients with greater amounts of exophoria at distance tend to have greater amounts of measured fusional divergence at near, while patients with esophoria at distance tend to have smaller amounts of prism divergence at near. Thus, the slow dissipation of phoria adaptation is indistinguishable from the measurement of fusional divergence amplitudes.

10) **Spread of comitance**: Although we generally attribute spread of comitance to extraocular muscle contracture, phoria adaptation can initiate and perpetuate spread of comitance long before extraocular muscle contracture develops.[22,25,26,31] For example, Kolling et al. found that prolonged occlusion restored incomitance in patients with superior oblique palsy and spread of comitance.[62] Similarly, the esodeviation in sixth nerve palsy may become more comitant over time, a phenomenon sometimes attributed to ipsilateral medial rectus contracture. However, medial rectus contracture should produce a spread of incomitance. So this paradoxical spread of comitance is more likely attributable to phoria adaptation, which can vary in different fields of gaze. As emphasized by Guyton,[20] phoria adaptation can eventuate in muscle length adaptation and secondary anatomical extraocular muscle contracture.

Neural substrate

Phoria adaptation has been considered by some,[10,63] but not others,[64] to be a cerebellar response, but midbrain vergence-related neurons may also play a role.[65] Visual adaptations to reversing prisms in the cat are preempted by lesions in the vestibulocerebellum.[66] In monkeys, removal of the cerebellar vermis (but not the flocculus) has been shown to disrupt phoria adaptation.[67,68]

When prisms are placed before one eye of an orthotropic patient, the visual image discordance is interpreted as an error signal. Adaptation and learning to correct motor error signals is modulated at the level of the vestibulocerebellum.[10,69] It, therefore, seems likely that phoria adaptation is similarly mediated by both climbing and mossy fibers within the cerebellum, which implement modifiable, adaptive, and “plastic” responses[10] (Figure 3).[70] The cerebellum receives continuous information via the mossy fiber system. The climbing fiber system originates from the inferior olivary nucleus and provides a powerful timing and error signal to Purkinje cells. The inferior olivary nucleus acts as a comparator of motor commands from the cerebral cortex, brainstem nuclei and receives feedback from receptors via the spinal cord, visual system, or vestibular organs. The inferior olive senses the error and recalibrates the tonic firing of the Purkinje cells. The increased frequency of inferior olivary nucleus discharge and complex spikes in the Purkinje cells triggers long-term depression of the synapse between the parallel fibers and the Purkinje cells, thereby resetting the single spike discharge rate to produce the necessary motor learning and adaptation. The Purkinje cell provides profound inhibition via GABA to the cerebellar nuclei, which provide the output of the cerebellum. It is therefore likely that the neural circuitry subserving phoria adaptation is not localized to a specific area but modulated by cortical, midbrain, and cerebellar circuitry.[10,65,66]

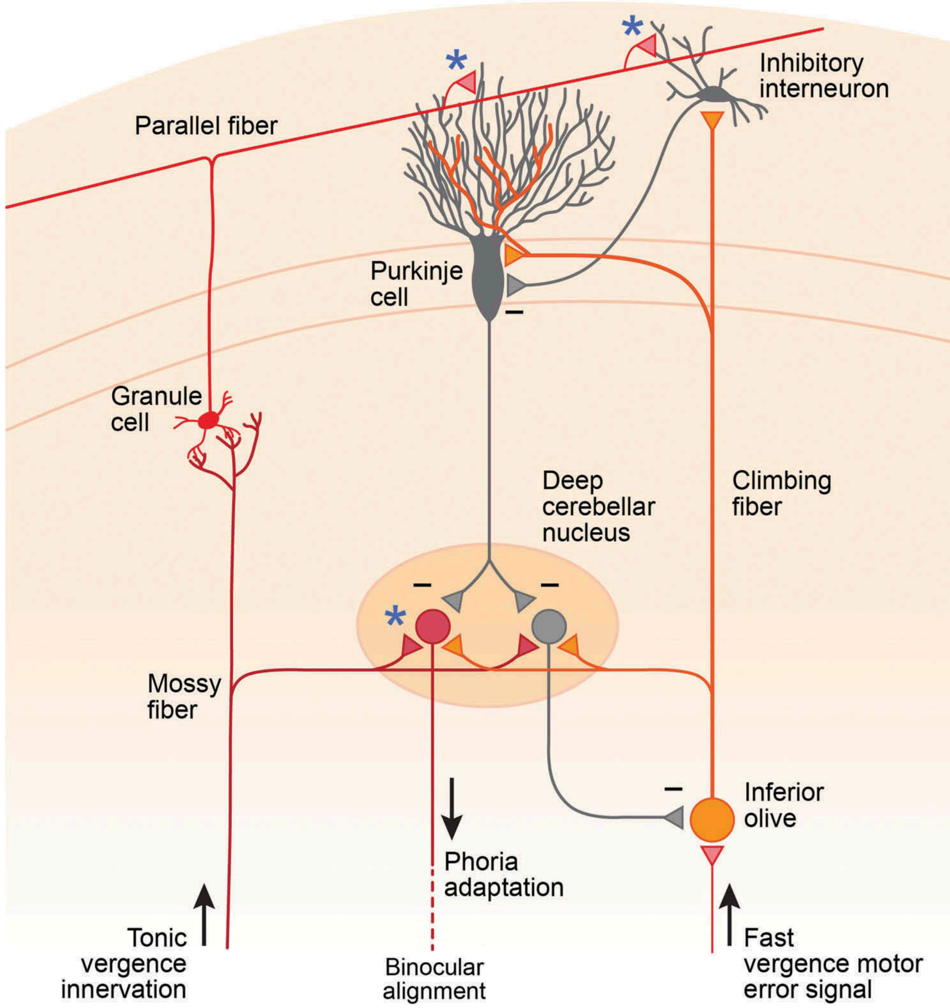

Figure 3. Diagram depicting cerebellar candidate pathways for the modulation of phoria adaptation. Complex spikes from mossy fibers transmit ongoing vergence information from the two eyes while slow simple spikes from climbing fibers input through inferior olive provide the error signal to modulate Purkinje cell output to the deep cerebellar nuclei and generate phoria adaptation (modified from reference 70, with permission).

Future questions

Is there a critical time period for the development of phoria adaptation? Could deficient phoria adaptation (due to a leaky neural integrator) engender some common forms of strabismus?[6,8,71,72] Can hemiretinal suppression impede phoria adaptation to allow intermittent deviations to perpetuate with time?[25] Since horizontal phoria adaptation is stronger in the convergent direction,[6,9,73] could phoria adaptation function as an on-off system rather than a push-pull system? Accordingly, could accommodative esotropia reflect a state of augmented phoria adaptation (driven by the accommodative cross-linkage instead of by vergence input), and intermittent exotropia reflect a state of deficient phoria adaptation? Do measured horizontal fusional divergence amplitudes simply represent the dissipation of horizontal phoria adaptation rather than the effects of an active divergence mechanism? Is deficient phoria adaptation associated with specific symptomatology?[8] Conversely, could some patients grow sensorially accustomed to a high level of phoria adaptation and feel subjectively uncomfortable when strabismus surgery removes the need for it? Do prism adaptation and prolonged monocular occlusion ultimately uncover the effects of phoria adaptation to the same extent? Should strabismus surgery be calibrated according to the measured phoria or to the deviation produced by the induced phoria adaptation? Does phoria adaptation supersede dissociated esotonus following surgical realignment of the eyes? Are the horizontal, vertical, and torsional components of phoria adaptation[25] anatomically segregated but mediated by the same final common pathways? Can a focal neurological lesion selectively disable a single component of phoria adaptation?

Conclusions

Phoria adaptation provides the central tonus mechanism that recalibrates binocular alignment to minimize the demands on fusional vergence, and thereby promote sensorimotor fusion. It is generated by a central neural integrator, probably within the cerebellum, that provides inertia, stability, and plasticity to human binocular vision. Although this shadow control system is often considered synonymous with "tonic vergence," phoria adaptation generates a deeper level of perseverative tonicity to maintain baseline extraocular muscle tonus so that sensorimotor fusion can be quickly and accurately superimposed. Clinically, its activation cannot readily be distinguished from its dissipation. Phoria adaptation is deeply entrenched before most of our prism measurements take place, making its effects virtually invisible to the clinician. As it so often adulterates our clinical strabismus measurements, it is truly the ghost in the machine.

As the effects of phoria adaptation may linger even after the largest possible phoria is measured,[74] this analysis begs the question of whether a "true phoria" can even exist,[17] or whether all measured phorias inevitably incorporate some degree of phoria adaptation. The ubiquity and robustness of this cerebellar learning mechanism in frontal binocular animals, and its augmentation in acquired forms of intermittent strabismus, would explain why our measured deviations so often underestimate the larger deviations that are obtained after prolonged occlusion or prism adaptation. When the effects of phoria adaptation fly under our radar, the resulting clinical measurements are often artificially low, and surgical undercorrections can be predicted. Thus, phoria adaptation may help to explain the trend toward progressively higher surgical dosing over the last century in intermittent exotropia and accommodative esotropia. Conversely, the malleability and plasticity of this shadow control system due to its inherent ability to "learn" may also explain why strabismus surgery is so often successful.[75]

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